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NPS/002/022 – Technical Specification for 66kV Power Cables

1. Purpose

The purpose of this document is to detail the technical requirements of CE Electric UK in relation to 66kV power cables.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/022	1 (June 06)	Technical Specification for 66kV Power Cables

2. Scope

This specification details the requirements for 66 kV power cables for use on CE Electric UK distribution network.

3. Technical Requirements

3.1 Conditions of Installation.

Cables specified in this document shall be installed in accordance with the CE Electric UK policy for the installation of cables: DSS/026. They will be pulled or laid into open trenches, pulled into ducts or installed in air. Cables may also be installed directly by trenchless installation techniques.

During storage and after installation cables can be expected to be subjected to the full range of climatic conditions encountered in the UK.

Cables may be surrounded by ground water for most of their operating lives. Where cables are installed in ducts, flooding of ducts can occur resulting in permanently wet sections along the cable route.

Cables installed above ground will be supported by means of cleats either vertically or horizontally and these cables may be exposed to direct sunlight for significant periods.

Cables may be installed up wooden structures and therefore in contact preservation medium such as creosote.

Accessories may be cold applied or require the application of heat.

3.2 Conditions of Operation

Power cables purchased in accordance with this specification are required to operate under conditions stipulated in IMP/001/913. The following are general conditions of operation:

- Nominal system voltages: 66 kV.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 6%.
- Nominal system frequency: 50Hz
- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points.

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- Suitable for 3 ph symmetrical short circuit currents of 20kA for 1 sec and earth fault currents of 3kA for 2 secs (**note:** at 66kV phase to earth currents are less than 3 ph currents) where the system has been designed with an earth fault factor of 1.73

3.3 66 kV Power Cables

66 kV power cables of single core extruded insulation design shall be in single core configuration only.

Cables shall comply in all respects with the testing requirements of BS 7912, IEC 885-2, and IEC 60840.

3.3.1 Conductors

Conductors shall be compacted stranded copper in accordance with BS EN 60228. The range of conductor cross sections and materials required is as detailed in Appendix 1.

3.3.2 Insulation and Screens

The conductor screen, insulation and insulation screen shall be applied as a continuous single pass triple extrusion free of factory repairs. The cross linking process will completely "Dry Cured" and no water will be used during this process. The vulcanisation tube shall be filled with either dry nitrogen or with oil that is compatible with the insulation and screen materials.

The nominal thickness of the semi conducting screens shall be 1mm with a minimum of 0.6mm at any one point around the circumference.

The nominal insulation thickness shall be 12mm.

The insulation screen shall be the "fully bonded" type.

All cables will be "De-gassed" for four days at a continual 65 ° C.

3.3.3 Lead Sheath and Moisture Blocking

All cables shall have a lead sheath which will normally comply with the required fault rating and no less than the source substation single phase to earth fault. However, in areas of high fault current, additional copper screen wires may be required.

All cables shall be blocked against longitudinal moisture transmission and shall pass the moisture penetration tests outlined in IEC 60840: 2004 (annexe F).

The lead sheath shall consist of a seamless tube of lead alloy extruded over the longitudinal moisture barrier

3.3.4 Over sheath

The over sheath shall consist of an extruded layer of black MDPE complying with the requirements of BS 6234 and the relevant sections of BS 7870 part 4. The compound shall have a density within the range quoted in ENATS 09-17.

A continuous extruded semi conducting layer shall be applied upon the outer sheath to provide an electrode for sheath voltage tests. A "baked" graphite layer may be offered as an option.

The over sheath shall be indent printed or embossed with the following legend along two or more evenly spaced lines.

ELECTRIC CABLE 66000 VOLTS

Additional markings indicating the manufacturing reference, the year and month of manufacture, and metre length marking can be either indented or printed.

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4. References

4.1 International Documentation

The cables shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant

Reference	Title
IEC 60840	Power cables and accessories, for rated voltages 66kV to 132kV. Requirements and test methods.
BS EN 60228	Conductors in Insulated cables
BS 6234	Specification for polyethylene insulation and sheath of electric cables
BS 7870	LV and MV Polymeric insulated cables for use by distribution and generation utilities
BS 7912	Power cables and accessories, for rated voltages 66kV to 132kV. Requirements and test methods
ENATS 09-02	Specification for the supply, delivery and installation of power cables with operating voltages in the range 33kV to 400kV and associated auxiliary cables
Cenelec HD 605	Electrical Cables – Additional test methods
ENATS 09-17	Single core cables for use in substations having extruded insulations and rated voltages of 6350/11000 and 19000/33000 volts

4.2 Internal documentation

Reference	Title
IMP/001/913	CoP for the Economic Development of the EHV System
DSS/026	Policy for the Installation of Distribution Power Cables

5 Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	09/06/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	26/06/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Standards Engineer	G Hammel	11/06/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	29/06/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	11/06/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	11/06/2009

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Appendix 1 – Cable Types Utilised

66 kV Cables (Ref : IMP/001/913)

Single core

- 300mm² Circular Stranded Copper Conductor
- 500mm² Circular Stranded Copper Conductor

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Appendix 2 – Addendum to Supplier Requirements

For each power cable offered the Tenderer shall provide the following cable rating data: -

- Cable rating laid direct (Amps)
- Cable rating laid in a 150mm internal diameter duct (Amps)
- Cable rating in air (Amps)
- The 3 ph symmetrical short circuit and phase to earth short circuit rating
- Maximum dc resistance per phase conductor at 20⁰C (Ohms/km)
- Maximum ac resistance per phase conductor at maximum conductor temperature (Ohms/km)
- Star reactance at 50Hz (Ohms/km)
- Star capacitance at 50Hz (Ohms/km)
- Charging current per phase at normal voltage and frequency (mA/m)
- Zero sequence impedance $R_0 + jX_0$ (Ohms/km)
- Minimum dynamic bending radius (mm) (In Triplex and single core formations)
- Minimum static bending radius (mm)
- The minimum average insulation thickness
- The maximum electrical stress on the conductor screen and insulation screen
- Recommended pulling method and maximum pulling tension (kgF)

The following assumptions shall be made when quoting ratings: -

Single core cables laid direct or in air are in close trefoil, in ducts trefoil at 150mm centres.

Cover to top of cables being 910mm

Ground Thermal Resistivity = 1.2 K m/W

Ground temperature = 15⁰C

Air temperature = 25⁰C

The oversheath of all cables shall be marked in accordance with the requirements of the relevant Standard specified in this document.

All cables shall be metre marked throughout the length of the cable and the start and end values shall be marked on the drum label.

Cables shall be marked with some form of reference or batch number that can be used to ensure batch trace ability of materials and manufacturing facilities used in the construction of the cable.

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Appendix 3 – Logistical Requirements

All cable drums shall be returnable and the tenderer shall arrange to collect empty drums from the company's normal delivery locations.

Tenderers shall state at the time of tender their proposed cable drum sizes and weights for each cable type offered.

All cable drums shall be marked in accordance with the relevant cable Specification or Standard. The drum label shall also contain

- (a) Customers catalogue number
- (b) Name of manufacturer
- (c) Supplied length
- (d) Rated voltage
- (e) Number of cores
- (f) Size of conductor
- (g) Type of conductor material ("Cu" or "Al")
- (h) Abbreviated description of cable construction
- (i) Gross and nett weights
- (j) Direction of rolling drum
- (k) The metre marking starts and end values
- (l) The unique reference number

Cable drums may be stored for long period's outdoors. All drum labels shall remain legible and durable under these conditions.

The ends of all cables shall be effectively sealed against the ingress of moisture by a method appropriate to the cable type. Tenderers shall detail at the time of tender their proposed sealing arrangement for each cable type offered.

The cable end projecting from the drum shall be protected from damage during transit, storage and handling on site.

The cable on the drum shall not be susceptible to damage during transit, storage and handling on site. Tenderers shall state at the time of tender their proposed method of protection for each cable

Each delivery length of cable shall be allocated a unique reference number. This number shall appear on the factory test sheet covering the cable length, shall be clearly marked on the drum on which the length is delivered and shall be referred to on all invoices and advice notes.

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

66 kV cables required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes, where appropriate indicating if tests are type or routine tests.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each type of cable offered and supplemented with a cable data sheet.

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		Clause / Requirements	Conformance Code	Remarks / Comments
	Conductor	BS EN 60228		
	Conductor Examination	IEC 60840 10.4		
	Conductor Resistance	IEC 60840 10.5		
	Conductor and Insulation Screen Thickness	BS 7912 9.6.3		
	Conductor Screen, Insulation & Insulation Screen:	Continuous single pass triple extrusion (free from factory repairs)		
		Fully Bonded		
		Dry Cured Cross Linking		
	Insulation Thickness	Nominal 12mm		
		IEC 60840 10.6.2 / BS 7912 9.6.2 & 12.4		
	Metallic Sheath Thickness	IEC 60840 10.7 / BS7912 9.6.4		
		Fault Current Capacity		
	Moisture Penetration Test	IEC 60840 (annexe F)		
	Oversheaths :	Extruded MDPE to BS 6234, BS 7870 Part 4 & Density as ENATS 09:17		
	- Electrical Test	IEC 60840 9.4		
	- Thickness	BS 7912 9.6.5		

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		Clause / Requirements	Conformance Code	Remarks / Comments
	Bending Test	BS 7912 12.5		
	Hot Set Test	IEC 60840 10.9		
	Partial Discharge Test	IEC 60840 9.2 / BS 7912 12.6		
	Voltage Test	IEC 60840 9.3		
	Voltage Test	BS 7912 12.10		
	Load Cycle Voltage Test	BS 7912 12.9		
	Impulse Voltage Test	BS 7912 12.11		